

ADVANCED PYTHON — DAY 10

Static & Class Methods

🕒 Duration: 2 Hours

Think of a School System

OBJECT



Instance Method

"Mera naam Rahul hai"

Works on a specific Student.

CLASS



Class Method

"Total 500 students hain"

Works on the School (Shared).

INDEPENDENT



Static Method

"2 + 2 = 4"

General utility logic.

Instance Method

- ▶ Works on **object data** directly.
- ▶ Uses the `self` parameter implicitly.
- ▶ Called using an object instance.

```
def greet(self):  
    print(f"Mera naam {self.name} hai")
```

```
57 for j in range(3):  
58     col = -1  
59     while col in triedCo  
60         col = randint(0,2)  
61         triedCol.append(col)  
62         if " " == board[block  
63         board[block][row]  
64         founds
```


Class Method

@classmethod

- ▶ Works on **class data** (variables).
- ▶ Uses the `cls` parameter instead of `self`.
- ▶ Data and logic is **shared** across all objects.

Example Code

```
@classmethod  
def get_school_info(cls):  
    print(cls.school_name)
```


Static Method

@staticmethod

- Takes no `self`, no `cls` parameters.
- Functions as an independent utility.
- No relation to object or class data.

```
@staticmethod
def is_pass(marks):
    return marks ≥ 35
```



Key Differences

Feature	Instance Method	Class Method	Static Method
Parameter	<code>self</code>	<code>cls</code>	<code>none</code>
Works on	Object data	Class data	Independent logic
Call Syntax	<code>obj.method()</code>	<code>Class.method()</code>	<code>Class.method()</code>

Practice: Bank Example

OBJECT DATA



Instance

`show_balance()`

Shows specific account balance
for a single user.

CLASS VARIABLES



Class

`bank_info()`

Retrieves overarching details
like total bank branches.

UTILITY LOGIC



Static

`is_valid_deposit()`

Checks if deposit amount is
greater than zero.

Practice Task: Employee Class



Your Goal

Create an `Employee` class covering all three method types. Think critically about what data belongs to the company versus the individual employee.

Implementation Checklist

- ▶ **Instance Method:** Print the employee's specific salary.
- ▶ **Class Method:** Print company name + total employee count.
- ▶ **Static Method:** Check if a salary is valid (e.g., amount > 0).

Quick Revision



Question 1

`@classmethod` ka first parameter kya hota hai?



Question 2

Static method bina object ke call ho sakta hai?



Question 3

Class variable access kaun karta hai?

Day 10 Completion Checklist

- ✓ Instance vs Class vs Static samjha
- ✓ Decorators (`@classmethod` , `@staticmethod`) use kiye
- ✓ Bank example practice kiya
- ✓ Employee class khud likha
- ✓ Revision complete

Image Sources



https://png.pngtree.com/thumb_back/fw800/background/20221015/pngtree-abstract-depiction-of-programming-workflow-on-a-screen-during-actual-python-code-development-photo-image_28458258.jpg

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